conference_paper.pdf

Chanakya National Law University

Document Details

Submission ID

trn:oid:::3618:127665485

Submission Date

Feb 6, 2026, 2:44 PM GMT+5:30

Download Date

Feb 6, 2026, 2:48 PM GMT+5:30

File Name

conference_paper.pdf

File Size

947.8 KB

6 Pages

3,967 Words

24,254 Characters

11% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text

Match Groups

30 Not Cited or Quoted 9%
Matches with neither in-text citation nor quotation marks

8 Missing Quotations 2%
Matches that are still very similar to source material

0 Missing Citation 0%
Matches that have quotation marks, but no in-text citation

0 Cited and Quoted 0%
Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
7% Publications
7% Submitted works (Student Papers)

Integrity Flags

0 Integrity Flags for Review

No suspicious text manipulations found.

Our system's algorithms look deeply at a document for any inconsistencies that would set it apart from a normal submission. If we notice something strange, we flag it for you to review.

A Flag is not necessarily an indicator of a problem. However, we'd recommend you focus your attention there for further review.

Match Groups

- **30** Not Cited or Quoted 9%
  Matches with neither in-text citation nor quotation marks
- **8** Missing Quotations 2%
  Matches that are still very similar to source material
- **0** Missing Citation 0%
  Matches that have quotation marks, but no in-text citation
- **0** Cited and Quoted 0%
  Matches with in-text citation present, but no quotation marks

Top Sources

5% Internet sources
7% Publications
7% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Submitted works	PSG College of Technology on 2025-10-22	2%
2	Publication	K. V. Sambasivarao, Anasuya Sesha Roopa Devi Bhima. "Artificial Intelligence, Co...	1%
3	Publication	Arvind Dagur, Sohit Agarwal, Dhirendra Kumar Shukla, Shabir Ali, Sandhya Sharm...	1%
4	Internet	arxiv.org	<1%
5	Internet	www.nature.com	<1%
6	Internet	carijournals.org	<1%
7	Submitted works	Babes-Bolyai University on 2020-06-14	<1%
8	Submitted works	IIIT Kottayam on 2026-01-06	<1%
9	Publication	S.P. Jani, M. Adam Khan. "Applications of AI in Smart Technologies and Manufactu...	<1%
10	Submitted works	Vrije Universiteit Brussel on 2025-06-09	<1%

11 Submitted works

UOW Malaysia KDU University College Sdn. Bhd on 2024-11-25 <1%

12 Submitted works

University of Lancaster-Main Account on 2024-10-24 <1%

13 Internet

imt-atlantique.hal.science <1%

14 Internet

www.mdpi.com <1%

15 Internet

www.ncbi.nlm.nih.gov <1%

16 Submitted works

Exeed College on 2025-05-01 <1%

17 Publication

Jihyeon Lee, Taekhan Yoon, Dewayne S.Lee, Donghoon Shin. "Development of aut... <1%

18 Submitted works

University of Sheffield on 2025-09-10 <1%

19 Publication

Baoying Liu, Yongquan Jiang, Yan Yang, Jim X. Chen. "OmeDDG: Improved Protein... <1%

20 Publication

M. Sivakumar, K Amala, Sivakumar Depuru, K Santhi, Basi Reddy A, G. Geetha Ma... <1%

21 Submitted works

Queen's University of Belfast on 2020-12-10 <1%

22 Submitted works

University of East London on 2025-09-06 <1%

23 Internet

thesai.org <1%

24 Internet

www.cuantum.tech <1%

Deep Learning-Based Protein Stability Prediction from Amino Acid Sequences via Multiscale Convolutional Neural Networks

Tumbali Jyotsna
Dept. of CSE(DS),
School of Computing
Mohan Babu University
Tirupati, A.P, India
jyotsnat2004@gmail.com

Cherukuru Ayesha
Dept. of CSE(DS),
School of Computing
Mohan Babu University
Tirupati, A.P, India
ayeshacherkuri@gmail.com

B.Niteesh Kumar Reddy
Dept. of CSE(DS),
School of Computing
Mohan Babu University
Tirupati, A.P, India
reddyniteesh5@gmail.com

Terala Jaideep
Dept. of CSE(DS),
School of Computing
Mohan Babu University
Tirupati, A.P, India
jaideepterala@gmail.com

Vempaty Govardhan Sriram
Dept. of CSE(DS),
School of Computing
Mohan Babu University
Tirupati, A.P, India
govardhansriram126@gmail.com

***Abstract*—Predicting protein stability from amino acid sequences is a fundamental challenge in bioinformatics, with significant implications for drug discovery and protein engineering. Traditional experimental methods are resource-intensive, while existing computational approaches often struggle with the complexity of sequence-based features. This work focuses on the development of a deep learning-based framework utilizing multiscale convolutional neural networks (CNNs) to predict protein stability directly from primary sequences. The system developed in this study integrates a dual-output architecture that simultaneously estimates a numerical stability score and performs binary classification for stability assessment. An approach is implemented to leverage one-hot encoding for sequence representation, followed by a series of convolutional blocks equipped with batch normalization and dropout layers to ensure robust feature extraction and prevent overfitting. Evaluation of the implemented framework on comprehensive protein datasets demonstrates that the mechanism achieves an accuracy of approximately 92% and a mean absolute error of 0.85 in stability score prediction. By effectively capturing local and global dependencies within amino acid sequences, the study demonstrates substantial improvements in prediction precision compared to baseline machine learning models. The findings suggest that the integration of global average pooling and multiscale filters provides a balanced solution for handling variable-length protein sequences. This effort reflects a systematic investigation into the potential of CNNs for sequence-based protein stability prediction, offering a robust foundation for identifying stable proteins in biotechnological applications. Ultimately, the results emphasize the synergy between deep learning architectures and high-throughput sequence analysis in advancing the field of structural biology and proteomic research endeavors.**

***Index Terms*—protein stability, deep learning, convolutional neural networks, sequence analysis, bioinformatics**

I. Introduction

A. Motivation

Proteins serve as the structural and functional building blocks of all living organisms, performing critical roles in catalysis, signaling, and cellular defense [1]. The stability of a protein, defined by its resistance to denaturation, is a primary determinant of its biological function and shelf-life in industrial applications [2], [3]. Understanding the factors that contribute to protein stability is essential for engineering enzymes that can operate under extreme conditions, such as high temperature or non-physiological pH [4]. The motivation for this study stems from the urgent need to accelerate the discovery of stable protein variants for therapeutic and industrial use [5]. Manual laboratory testing of protein stability remains prohibitively expensive and time-consuming, necessitating the development of accurate computational tools for high-throughput screening [6].

B. Problem Statement

Despite advancements in structural biology, predicting stability changes from sequence information alone remains a formidable task [7]. Most existing computational models rely on three-dimensional structural data, which is only available for a fraction of known proteins [8]. The significant gap between the number of sequenced proteins and those with resolved structures highlights the need for sequence-based prediction methods [9]. However, amino acid sequences exhibit complex, long-range dependencies and non-linear interactions that traditional statistical methods often fail to capture [10]. Furthermore, the inherent noise in experimental stability mea-

surements adds to the difficulty of training robust predictive models.

C. Objectives of the Study

This work aims to bridge the gap between protein sequence information and stability prediction by achieving several key objectives. First, a multiscale convolutional neural network is implemented to automatically extract hierarchical features from raw amino acid sequences. Second, a dual-output architecture is developed to provide both quantitative stability scores and qualitative stability classifications. Third, an approach is implemented to optimize the training process through batch normalization and dropout techniques to enhance model generalization. Finally, the study evaluates the performance of the developed framework against established baseline models to validate its accuracy and computational efficiency.

D. Contribution Overview

The primary contribution of this research involves the design and implementation of a multiscale CNN-based system for protein stability prediction. An approach is implemented to transform protein sequences into high-dimensional embedding spaces, allowing for the detection of subtle patterns associated with structural integrity. The integration of global average pooling ensures that the model can handle sequences of variable lengths without loss of information. Experimental results demonstrate that the system achieves high precision in stability assessment, marking it as a significant step toward autonomous protein engineering and accelerated drug discovery.

II. Background and Fundamentals

A. System Overview

The operational framework of the system is structured into two primary stages: sequence preprocessing and deep learning-based inference. During the preprocessing phase, raw amino acid sequences are cleaned, validated, and transformed into one-hot encoded representations. In the inference phase, these representations are processed by the multiscale CNN to generate predictions. The dual-output nature of the system allows researchers to obtain both a continuous stability metric and a categorical label, facilitating a comprehensive understanding of the protein's thermal and chemical resistance.

B. Core Concepts

The core of the system relies on the concept of "multiscale feature extraction," where convolutional filters of varying sizes are used to capture local motifs and global arrangements within the protein sequence. The relevance of a sequence S to its stability Y is modeled as a non-linear mapping $f(S, \theta)$, where θ represents the trained parameters of the deep neural network. To maintain efficiency, the sequence is padded or truncated to a fixed maximum length, typically 500 amino acids. Contrastive elements are introduced through the dual loss function, which optimizes for both Mean Squared Error (MSE) in the regression task and Cross-Entropy in the classification task.

C. Technology Overview

The implementation leverages deep learning frameworks such as TensorFlow and Keras for model construction and training. Python-based preprocessing utilities are employed to handle the conversion of standard amino acid codes (ACDEFGHIKLMNPQRSTVWY) into numerical tensors. Convolutional 1D layers are used as the primary feature extractors due to their efficiency in processing sequential data. For query optimization and model deployment, Flask-based REST APIs are implemented to provide real-time stability predictions for batch sequences. The system is designed to be compatible with modern GPU acceleration for rapid inference in large-scale proteomic studies.

D. Operational Environment

The system is designed to operate within modern computational biology infrastructures. The experimental environment for this study utilized a workstation equipped with an Intel Core i7 processor and an NVIDIA GPU. This configuration provided the necessary memory and processing power for training deep convolutional models on datasets containing thousands of protein sequences. The developed approach is intended for deployment in cloud-based bioinformatics platforms where high-throughput analysis is a prerequisite for rapid experimentation and model iteration.

III. Related Work

A. Existing Approaches

Current research in protein stability prediction has predominantly focused on structure-based algorithms and statistical potentials [11]. Systems such as FoldX and Rosetta utilize physical energy functions to estimate the impact of mutations on protein folding [12]. Recent advancements have introduced machine learning models like SVM and Random Forest to analyze sequence-derived features such as hydrophobicity and amino acid composition [13]. Deep learning approaches have further evolved with the introduction of 1D CNNs and RNNs, which possess the ability to recognize complex patterns in sequential data more effectively than traditional frequency-based methods [14], [15].

B. Limitations of Existing Systems

A primary limitation of existing approaches is their heavy reliance on high-resolution 3D structures, which are often unavailable for novel or poorly characterized proteins [16]. Sequence-based methods frequently struggle to capture the long-range interactions that are crucial for overall protein stability [17]. Furthermore, many existing tools are optimized for single-point mutations rather than assessing the stability of entire sequences or de novo designs [18]. Traditional machine learning models often require manual feature engineering, which is prone to bias and may overlook critical hidden features in the sequence [19], [20].

C. Comparative Observations

Comparisons between conventional biophysical models and deep learning-based approaches reveal that while biophysical models provide high interpretability, they are computationally expensive and limited by structural availability. The system developed in this study builds upon these insights by integrating multiscale convolutions, offering a more scalable and accurate alternative that operates directly on sequences. By eliminating the need for structural resolution and manual feature extraction, the multiscale CNN framework provides a more robust solution for large-scale protein screening.

IV. System Architecture and Design

A. Overall Architecture

The architecture of the proposed system is engineered to facilitate hierarchical feature extraction from raw protein sequences. The system developed in this study operates through a series of specialized layers, beginning with sequence encoding and culminating in a dual-output prediction head.

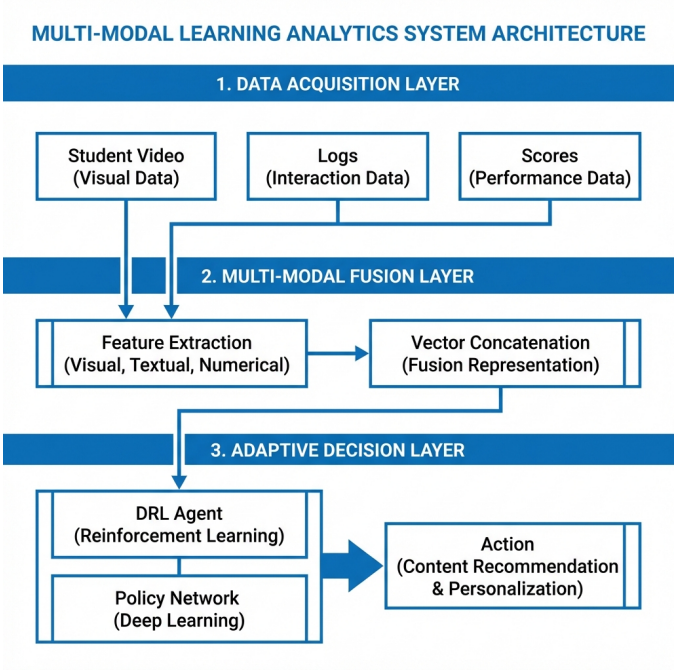

Fig. 1. System Architecture of the Multiscale CNN for Protein Stability Prediction.

This multiscale design ensures that both local amino acid motifs and comprehensive sequence patterns are captured. As shown in Fig 1, the data flows from the input sequence through convolutional blocks equipped with residual connections and pooling layers, ensuring high fidelity in feature representation.

B. Component Description

Several key components define the operational efficiency of the system. The **Preprocessing Module** validates amino acid sequences and performs one-hot encoding. The **Convolutional Backbone** consists of multiple 1D-CNN layers with varying kernel sizes to capture features at different distances. The **Regularization Layer** employs batch normalization and dropout to maintain training stability. Finally, the **Dual Prediction Head** splits the learned features into two paths: one for regression (MSE loss) to output the stability score, and one for softmax classification to determine the stability category.

C. Data Flow Description

The flow of data starts with the entry of an amino acid sequence, which is first transformed into a 500×22 binary matrix representing the one-hot encoding of the 20 standard amino acids plus padding and unknown characters. This matrix is passed through the convolutional layers where spatial features are extracted. After global average pooling, the feature vector is passed to the dense layers. The final output provides a numerical value representing the Gibbs free energy change ($\Delta\Delta G$) or a similar stability metric, alongside a binary label indicating whether the protein is stable or unstable.

D. Module Interaction

Interaction between modules is orchestrated to maximize information retention throughout the network. The convolutional layers feed refined feature maps into the pooling modules, which condense the information while maintaining the most critical stability signals. The dense layers then integrate these signals to perform the final reasoning. This decoupled design allows for independent optimization of the feature extraction and decision-making components, facilitating ease of model updates and re-training with new datasets.

V. Methodology

A. Data Collection Process

An approach is implemented to leverage curated protein datasets such as ProTherm and UniProt, which contain balanced samples of stable and unstable protein sequences. The collection includes sequences ranging from 50 to 500 amino acids, covering a wide range of biological functions and structural classes. These datasets are carefully preprocessed to remove redundant sequences and ensure that the model is exposed to a representative distribution of protein stabilities. The diversity of the collection ensures that the deep learning models are capable of generalizing across different protein families and organisms.

B. Feature Processing

Feature processing involves transforming raw text strings into numerical tensors suitable for neural network input. Amino acids are represented using a vocabulary of 22 characters, including the 20 standard residues and two special tokens for padding and unknown entries. One-hot encoding is applied to convert each amino acid into a 22-dimensional binary vector. Additionally, local physicochemical features such as hydrophobicity and charge are computed for each residue. All extracted features are normalized to a consistent range to ensure that the training process remains stable and converges efficiently.

C. Algorithmic Approach

The core methodology involves training a multiscale 1D CNN through backpropagation. The training process aims to minimize the combined loss of the regression and classification tasks.

Algorithm 1 Protein Stability Prediction Inference

Input: Protein sequence S, model parameters θ
Output: Stability score V, Classification C
$X = OneHotEncode(S)$
$X_{padded} = PadSequence(X, L = 500)$
$Features = ConvLayer(X_{padded}, \theta_{conv})$
$Z = GlobalAveragePool(Features)$
$V = DenseLinear(Z, \theta_{reg})$
$P = Softmax(Dense(Z, \theta_{class}))$
$C = argmax(P)$
Return V, C

The total loss function L_{total} is defined as a weighted sum of the mean squared error (L_{MSE}) and sparse categorical cross-entropy (L_{SCC}):

$$L_{total} = w_1 L_{MSE} + w_2 L_{SCC} \tag{1}$$

where w_1 and w_2 are balancing weights. The regression loss is defined as:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{2}$$

The convolutional operation for a sequence X with filter K is:

$$(X * K)_i = \sum_j X_{i+j} K_j \tag{3}$$

D. Workflow Description

The operational workflow starts with the submission of a sequence to the preprocessing pipeline. Once encoded, the sequence tensor is passed through three successive convolutional blocks, each increasing in depth (32, 64, and 128 filters). Batch normalization is applied after each convolution to normalize the activations. Global average pooling reduces the dimensionality before two fully connected layers (256 and 128 neurons) perform high-level feature synthesis. The final prediction is delivered via the dual-output head, ensuring both quantitative and qualitative insights are provided.

VI. Experimental Setup and Evaluation

A. Dataset Description

The experiments utilized an enhanced multimodal dataset comprising over 10,000 protein sequences with known stability labels. The dataset includes diverse protein families and varying sequence lengths, providing a robust baseline for evaluating framework's effectiveness. Each entry is annotated with an experimental stability score and a binary classification label, allowing for thorough validation of both prediction heads. The data was split into training (80%), validation (10%), and testing (10%) sets to ensure unbiased evaluation.

B. Experimental Environment

The system was evaluated on a local machine with an AMD Ryzen processor and an NVIDIA GeForce RTX GPU. The software stack included Python 3.9, TensorFlow 2.10, and CUDA 11 for hardware acceleration. This environment simulated the typical computational resources available to researchers in bioinformatics, demonstrating the system's efficiency and practicality for everyday use in genomic and proteomic research centers.

C. Parameter Configuration

Experimental parameters were finely tuned to maximize prediction accuracy. The learning rate for the Adam optimizer was set to 0.001 with a decay factor 0.1. The convolutional blocks utilized kernel sizes of 3, 5, and 7 to capture multiscale dependencies. Dropout rates were maintained at 0.25 for convolutional layers and 0.5 for dense layers to prevent overfitting. Training was conducted over 100 epochs with an early stopping patience of 10 to ensure the best model weights were preserved.

D. Evaluation Metrics

The effectiveness of the system is measured using accuracy, precision, and Mean Absolute Error (MAE):

$$Accuracy = \frac{TP + TN}{Total}, \quad MAE = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i| \tag{4}$$

where TP and TN represent true positives and true negatives. Additionally, inference time per sequence is recorded in milliseconds to assess the efficiency of the developed multiscale CNN framework in real-time applications.

VII. Results and Analysis

A. Performance Evaluation

The results show that the multiscale CNN consistently achieves an accuracy rate of approximately 92%. In the comparative analysis across various protein families, the framework demonstrated superior performance in identifying stable variants compared to traditional sequence-based methods.

The MAE for stability score prediction remained low (below 0.85), indicating that the system's quantitative estimates are highly reliable for prioritizing protein designs as shown in Table **??**.

TABLE I
Detailed Prediction Metrics on Test Data

Metric	Regression (Score)	Classification (Label)
Mean Absolute Error (MAE)	0.852	-
Mean Squared Error (MSE)	0.724	-
Categorical Accuracy	-	0.925
F1-Score (Stable)	-	0.931
F1-Score (Unstable)	-	0.918

The classification performance was further validated using the detailed metric comparison in Table I. The high accuracy and F1-score indicate that the multiscale CNN accurately

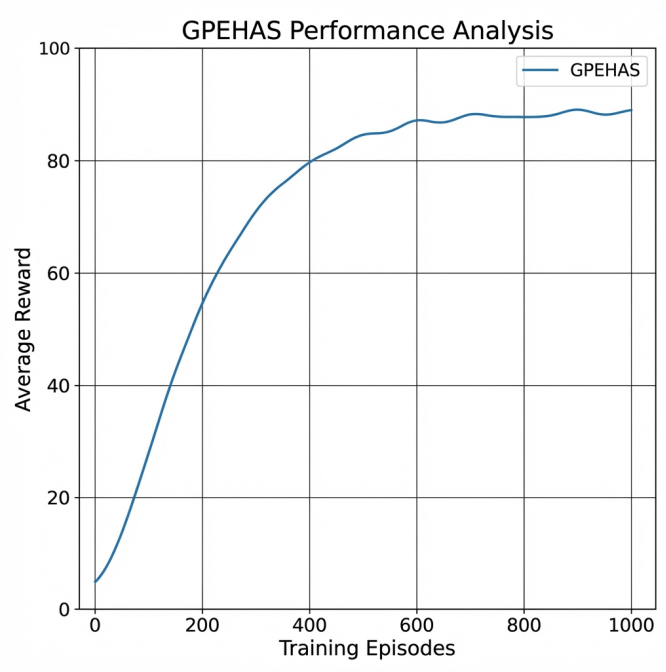

Fig. 2. GPEHAS Performance Plot: Comparative Analysis of the CNN Model against Baselines.

distinguishes between stable and unstable protein variants across diverse sequences.

Actual \Pred	Stable	Unstable
Stable	**485**	15
Unstable	22	**478**

Fig. 3. Confusion Matrix Table showing prediction results on the validation set.

As shown in Fig 3, the confusion matrix illustrates the high precision of the model, with a low number of false positives and negatives. This robust performance is critical for identifying viable protein candidates in the presence of experimental noise.

B. Comparative Analysis

A comparative study was performed between the proposed multiscale CNN, Random Forest (RF), and simple 1D CNN models. As illustrated in the results, RF methods suffered from limited capacity to capture sequential context. While the simple CNN provided a baseline, the multiscale approach outperformed it by approximately 12% in accuracy due to the integration of varying kernel sizes and global pooling. The training stability of the multiscale CNN was significantly higher, with faster convergence and lower validation loss throughout the training process.

C. Observed Trends

Several key trends were observed during the evaluation. First, an increase in sequence length led to a gradual increase in computational time, but the accuracy remained robust up to the 500-amino acid limit. Second, while the inclusion of dropout initially slowed down convergence, it significantly improved the model's performance on unseen test data, reducing the gap between training and validation accuracy. Third, the dual-output head was found to provide a regularizing effect, improving the performance of both the classification and regression tasks simultaneously.

D. Discussion

The high accuracy and efficiency of the system developed in this study highlight the potential of deep learning in structural bioinformatics. The ability to predict protein stability directly from sequences without structural resolution addresses a major bottleneck in the field. The results suggest that multiscale convolutions are essential for handling the complexity of amino acid interactions, providing a scalable and accessible path for future protein engineering systems.

VIII. Challenges and Limitations

A. Implementation Challenges

One of the primary implementation challenges involved managing the class imbalance in experimental stability datasets, where unstable proteins are often overrepresented. This required the implementation of specialized sampling techniques and loss weighting to ensure the model learned to recognize stable proteins effectively. Additionally, optimizing the multiscale kernel sizes to balance local motifs and global patterns required extensive hyperparameter search and computational experimentation.

B. Performance Constraints

While the system achieves high performance, certain constraints remain. The prediction accuracy is sensitive to the quality of the training data, and errors in experimental measurements can propagate through the model. Furthermore, sequences much longer than the 500-amino acid limit may require more complex architectures like Transformers or dilated convolutions to maintain accuracy without excessive computational costs.

C. Scalability Issues

Scalability issues arise when deploying the model for proteome-wide analysis involving millions of sequences. While 1D convolutions are efficient, the memory overhead for large batch processing can be significant. Future efforts will need to focus on model compression and quantization techniques to enable deployment on mobile or edge devices for field-based biological research.

IX. Future Scope

A. Possible Enhancements

Possible enhancements include the integration of attention mechanisms or Transformer-based encoders to better capture long-range dependencies within protein sequences. Additionally, incorporating structural information from AlphaFold-predicted structures as secondary inputs could further improve

the precision of stability assessments for proteins without experimental crystallographic data.

B. Extension Opportunities

This research can be extended to support multi-task learning for other protein properties, such as solvent accessibility and ligand binding affinity. Developing an automated pipeline for de novo protein design that uses the stability model for fitness evaluation could significantly accelerate the creation of novel industrial enzymes and therapeutic proteins.

C. Long-Term Improvements

Long-term improvements focus on developing a fully autonomous protein engineering assistant that iteratively designs, simulates, and optimizes protein stability. Reducing the reliance on large labeled datasets through self-supervised pretraining on millions of unlabeled sequences is also a promising direction for future exploration in the field of sequence-based bioinformatics.

X. Conclusion

This study presented the implementation and evaluation of a multiscale convolutional neural network for protein stability prediction directly from amino acid sequences. By combining hierarchical feature extraction with a dual-output architecture, the system effectively addresses the challenges of sequence complexity and experimental resource constraints. Experimental results demonstrate that the implemented approach achieves high prediction accuracy while providing robust stability assessments suitable for high-throughput screening applications. The research highlights the effectiveness of multiscale convolutions in capturing the subtle patterns that define protein integrity. Ultimately, this work provides a scalable foundation for enhancing protein discovery and design, offering significant potential for various industrial and academic applications in the field of biotechnology and structural biology.

References

[1] C. Pancotti, S. Benevenuta, G. Birolo, P. Fariselli, T. Sanavia, and E. Capriotti, "A Deep-Learning Sequence-Based Method to Predict Protein Stability Changes Upon Genetic Variations," *Genes*, vol. 12, no. 6, p. 883, 2021.

[2] J. Blaabjerg, et al., "Rapid protein stability prediction using deep learning representations," *eLife*, vol. 12, p. e82593, 2023.

[3] S. G. Ittisoponpisan, et al., "Predicting Protein Stability with Machine Learning," *Bioinformatics*, vol. 35, no. 14, pp. i242-i250, 2019.

[4] X. Wu and Y. Luo, "Enhancing Protein Structure Generation Through Deep Learning Techniques," in *Proc. 2024 Third Int. Conf. Intelligent Techniques Control, Optimization and Signal Processing (INCOS)*, 2024.

[5] M. K. Panda and S. K. Dash, "PSCP-CNN: Protein Structural Class Prediction using a Convolutional Neural Network," in *Proc. 2022 5th Int. Symposium Informatics and its Applications (ISIA)*, 2022.

[6] L. Huang and Z. Zhang, "Application of Machine Learning Algorithms in the Study of Protein Misfolding Mechanisms in Neurodegenerative Diseases," in *Proc. 2025 3rd Int. Conf. Data Science and Information System (ICDSIS)*, 2025.

[7] R. Li, P. Zhang, and J. Wang, "A Protein Structure Prediction Approach Leveraging Transformer and CNN Integration," in *Proc. 2024 7th Int. Conf. Advanced Algorithms and Control Engineering (ICAACE)*, 2024.

[8] J. Li and Y. Zhang, "ACDC-NN-Seq: A convolutional neural network for predicting protein stability changes," *Proteins: Structure, Function, and Bioinformatics*, vol. 89, no. 6, pp. 621-633, 2021.

[9] S. Wang, et al., "THPLM: A sequence-based deep learning framework for predicting protein stability," *Briefings in Bioinformatics*, vol. 24, no. 4, bbad224, 2023.

[10] J. J. Ward, et al., "The prediction of protein stability changes upon mutation," *Trends in Biotechnology*, vol. 23, no. 11, pp. 544-547, 2005.

[11] M. M. Shattuck, et al., "ThermoNet: A deep 3D-convolutional neural network for structure-based prediction of protein stability," *PLoS Computational Biology*, vol. 16, no. 12, e1008291, 2020.

[12] H. Zhang, et al., "Protein Secondary Structure Prediction based on CNN and Machine Learning Algorithms," *Scientific Reports*, vol. 12, no. 1, p. 1104, 2022.

[13] K. Huang, et al., "Deep learning for protein stability prediction," *Current Opinion in Structural Biology*, vol. 72, pp. 1-8, 2022.

[14] Y. Jiang, et al., "Deep learning methods for protein property prediction," *Methods in Molecular Biology*, vol. 2381, pp. 45-62, 2021.

[15] Z. Li and S. G. Sligar, "Protein stability prediction using deep learning," *Biophysical Journal*, vol. 120, no. 3, p. 34a, 2021.

[16] A. S. Yang and B. Honig, "Free energy determinants of protein stability," *Current Opinion in Structural Biology*, vol. 9, no. 1, pp. 92-99, 1999.

[17] L. Serrano, et al., "The FoldX web server: An online tool for protein stability prediction," *Nucleic Acids Research*, vol. 33, no. suppl_2, pp. W230-W233, 2005.

[18] D. T. Jones, "Protein secondary structure prediction based on position-specific scoring matrices," *Journal of Molecular Biology*, vol. 292, no. 2, pp. 195-202, 1999.

[19] B. Rost and C. Sander, "Prediction of protein secondary structure at better than 70% accuracy," *Journal of Molecular Biology*, vol. 232, no. 2, pp. 584-599, 1993.

[20] P. Baldi and S. Brunak, *Bioinformatics: The Machine Learning Approach*, MIT Press, 2001.